

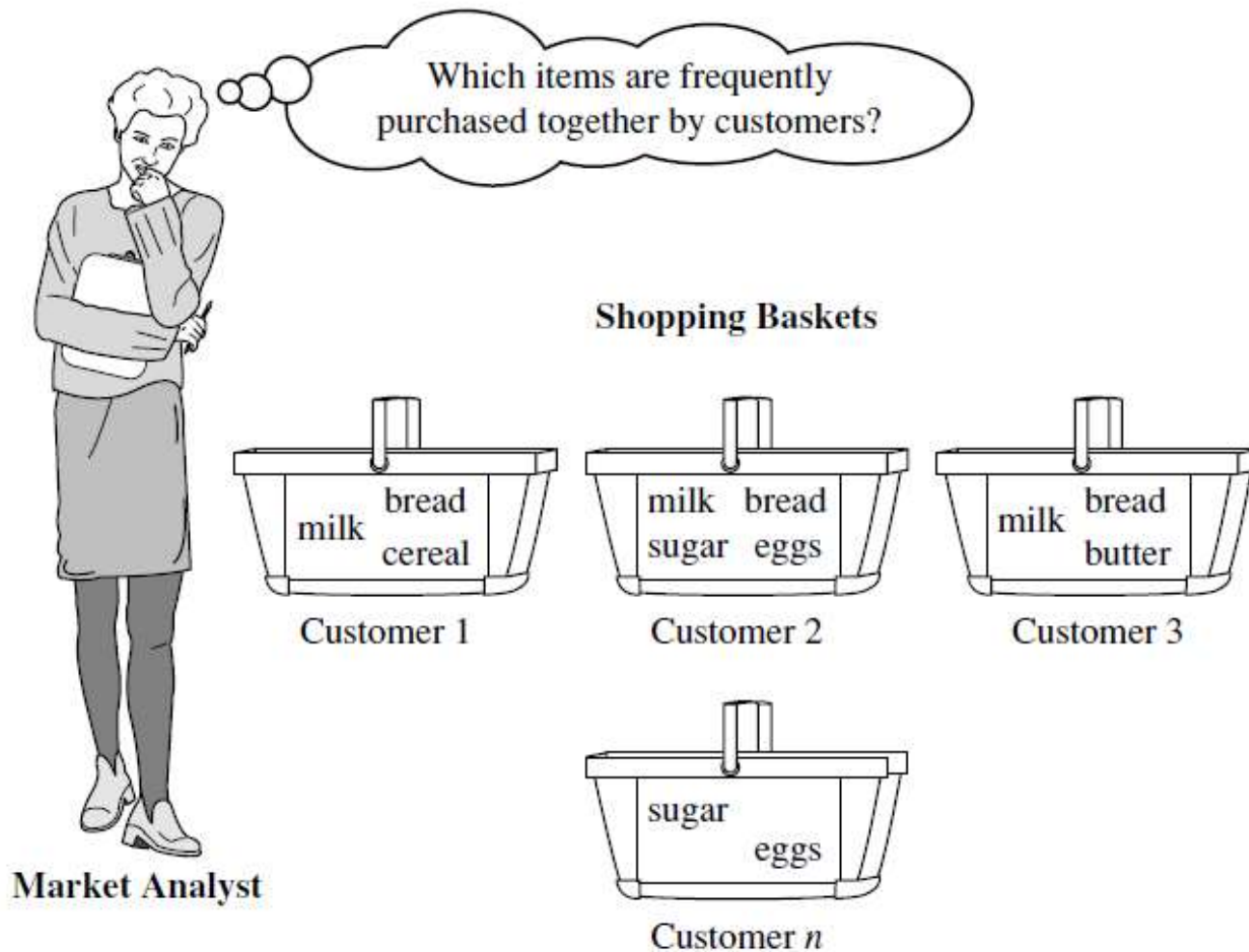
Unit3

Mining Frequent Patterns: Basic Concept – Frequent Item Set Mining
Methods -Apriori and FP Growth algorithms -Mining Association Rules

What Is Frequent Pattern Analysis?

- **Frequent pattern**: a pattern (a set of items, subsequences, substructures, etc.) that occurs frequently in a data set
- First proposed by Agrawal, Imielinski, and Swami [AIS93] in the context of **frequent itemsets** and **association rule mining**
- Motivation: Finding inherent regularities in data
 - What products were often purchased together?— Beer and diapers!
 - What are the subsequent purchases after buying a PC?
 - What kinds of DNA are sensitive to this new drug?
 - Can we automatically classify web documents?
- Applications
 - Basket data analysis, cross-marketing, catalog design, sale campaign analysis, Web log (click stream) analysis, and DNA sequence analysis.

Example : Market Basket Analysis



Example : Market Basket Analysis

- This process analyzes customer buying habits by finding associations between the different items that customers place in their “shopping baskets” .
- The discovery of these associations can help retailers develop marketing strategies by gaining insight into which items are frequently purchased together by customers.
- if customers are buying milk, how likely are they to also buy bread (and what kind of bread) on the same trip to the supermarket?
- This information can lead to increased sales by helping retailers do selective marketing and plan their shelf space.

Association rules

The customers who purchase computers also tend to buy antivirus software at the same time is represented in the following association rule:

computer $\Rightarrow$ *antivirus_software* [*support* = 2%, *confidence* = 60%].

Association rules

- Rule **support** and **confidence** are two measures of rule interestingness.
- They reflect the usefulness and certainty of discovered rules.
- A support of 2% for Rule means that 2% of all the transactions under analysis show that computer and antivirus software are purchased together.
- A confidence of 60% means that 60% of the customers who purchased a computer also bought the software.
- Typically, association rules are satisfy both a **minimum support threshold** and a **minimum confidence threshold**.

Support

- Support refers to **the relative frequency of an item set in a dataset.**
- For example, if an itemset occurs in 5% of the transactions in a dataset, it has a support of 5%.
- Support is often used as a threshold for identifying frequent item sets in a dataset, which can be used to generate association rules.
- **Support(X) =**
(Number of transactions containing X) / (Total number of transactions)
- For example, if we set the support threshold to 5%, then any itemset that occurs in more than 5% of the transactions in the dataset will be considered a frequent itemset.

- For example, suppose we have a dataset of 1000 transactions, and the itemset {milk, bread} appears in 100 of those transactions. The support of the itemset {milk, bread} would be calculated as follows:

$$\begin{aligned}\text{Support}(\{\text{milk, bread}\}) &= \frac{\text{Number of transactions containing } \{\text{milk, bread}\}}{\text{Total number of transactions}} \\ &= 100 / 1000 \\ &= 10\%\end{aligned}$$

Confidence

- Confidence is a measure of the reliability for a given association rule.
- It is defined as the proportion of cases in which the association rule holds true.
- For example: suppose we have a dataset of 1000 transactions, and the itemset {milk, bread} appears in 100 of those transactions.
- The itemset {milk} appears in 200 of those transactions. The confidence of the rule “If a customer buys milk, they will also buy bread” would be calculated as follows:
- $\text{Confidence}(X \Rightarrow Y) = (\text{Number of transactions containing } X \text{ and } Y) / (\text{Number of transactions containing } X)$

Confidence("If a customer buys milk, they will also buy bread")

= Number of transactions containing
{milk, bread} / Number of transactions
containing {milk}

= 100 / 200

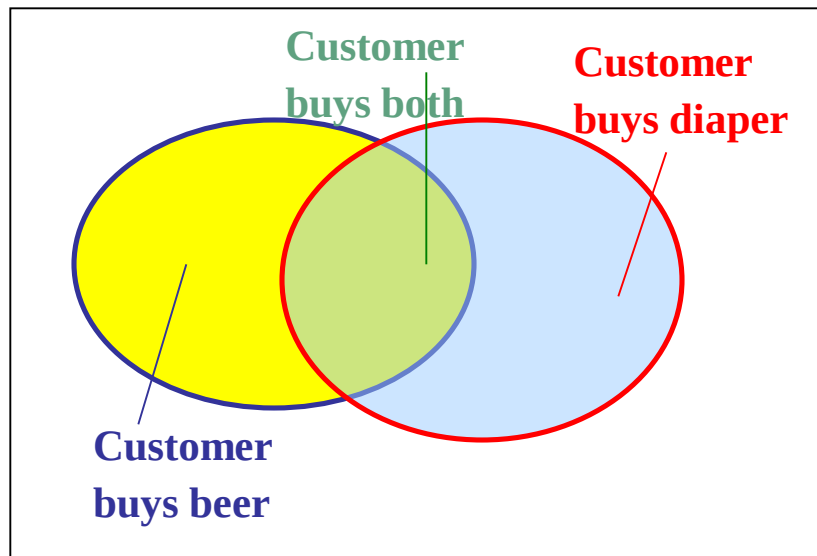
= 50%

Support and Confidence

Support	Confidence
Support is a measure of the number of times an item set appears in a dataset.	Confidence is a measure of the likelihood that an itemset will appear if another itemset appears.
calculated by dividing the number of transactions containing an item set by the total number of transactions.	calculated by dividing the number of transactions containing both itemsets by the number of transactions containing the first itemset.
It is used to identify itemsets that occur frequently in the dataset.	It is used to evaluate the strength of a rule.
it is often used with a threshold to identify itemsets that occur frequently enough to be of interest.	often used with a threshold to identify rules that are strong enough to be of interest.
Support is interpreted as the percentage of transactions in which an item set appears.	Confidence is interpreted as the percentage of transactions in which the second itemset appears given that the first itemset appears.

Basic Concepts: Frequent Patterns

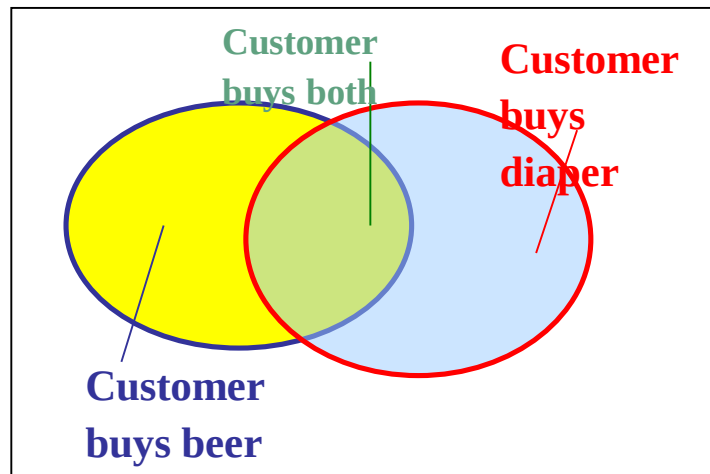
Tid	Items bought
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diaper
30	Beer, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk



- **itemset**: A set of one or more items
- **k-itemset** $X = \{x_1, \dots, x_k\}$
- **(absolute) support**, or, **support count** of X : Frequency or occurrence of an itemset X
- **(relative) support**, s , is the fraction of transactions that contains X (i.e., the **probability** that a transaction contains X)
- An itemset X is **frequent** if X 's support is no less than a minsup threshold

Basic Concepts: Association Rules

Tid	Items bought
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diaper
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40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk



- Find all the rules $X \rightarrow Y$ with minimum support and confidence
 - support**, s , **probability** that a transaction contains $X \cup Y$
 - confidence**, c , **conditional probability** that a transaction having X also contains Y

Let $minsup = 50\%$, $minconf = 50\%$

Freq. Pat.: Beer:3, Nuts:3, Diaper:4, Eggs:3, {Beer, Diaper}:3

- Association rules: (many more!)
 - Beer $\rightarrow$ Diaper (60%, 100%)
 - Diaper $\rightarrow$ Beer (60%, 75%)

Example: Association rule

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

-
- Support Count– Frequency of occurrence of a itemset.
 - Here {Milk, Bread, Diaper}=2
 - Frequent Itemset – An itemset whose support is greater than or equal to minsup threshold.
 - Association Rule – An implication expression of the form $X \rightarrow Y$, where X and Y are any 2 item sets.
 - Example: {Milk, Diaper} $\rightarrow$ {Beer}

Rule Evaluation Metrics

Support(s) –The number of transactions that include items in the $\{X\}$ and $\{Y\}$ parts of the rule as a percentage of the total number of transaction.

- It is a measure of how frequently the collection of items occur together as a percentage of all transactions.
- $\text{Support} = \sigma(X+Y) \backslash \text{total}$
- It is interpreted as fraction of transactions that contain both X and Y.

Confidence(c) –

- It is the ratio of the no of transactions that includes all items in $\{B\}$ as well as the no of transactions that includes all items in $\{A\}$ to the no of transactions that includes all items in $\{A\}$.
- $\text{Conf}(X \Rightarrow Y) = \text{Supp}(XUY) \backslash \text{Supp}(X)$
- It measures how often each item in Y appears in transactions that contains items in X also.

Example – From the above table, $\{\text{Milk, Diaper}\} \Rightarrow \{\text{Beer}\}$

$$\begin{aligned} s &= \\ \sigma & \\ (\{\text{Milk, Diaper, Beer}\}) & \\ \div & \\ |T| & \\ = 2/5 & \\ = 0.4 & \end{aligned}$$

$$\begin{aligned} c &= \\ \sigma & \\ (\text{Milk, Diaper, Beer}) & \\ \div & \end{aligned}$$

$$\begin{aligned} \sigma & \\ (\text{Milk, Diaper}) & \\ = 2/3 & \\ = 0.67 & \end{aligned}$$

$$\begin{aligned} l &= \text{Supp}(\{\text{Milk, Diaper, Beer}\}) \\ \div & \\ \text{Supp}(\{\text{Milk, Diaper}\}) * \text{Supp}(\{\text{Beer}\}) & \\ = 0.4 / (0.6 * 0.6) & \\ = 1.11 & \end{aligned}$$

Apriori: A Candidate Generation & Test Approach

- Apriori pruning principle: If there is **any** itemset which is infrequent, its superset should not be generated/tested! (Agrawal & Srikant @VLDB'94, Mannila, et al. @ KDD' 94)
- Method:
 - Initially, scan DB once to get frequent 1-itemset
 - **Generate** length $(k+1)$ **candidate** itemsets from length k **frequent** itemsets
 - **Test** the candidates against DB
 - Terminate when no frequent or candidate set can be generated

Steps in the Apriori Algorithm

- Step 1: Generate Frequent Itemset
- Step 2: Generate Association Rules

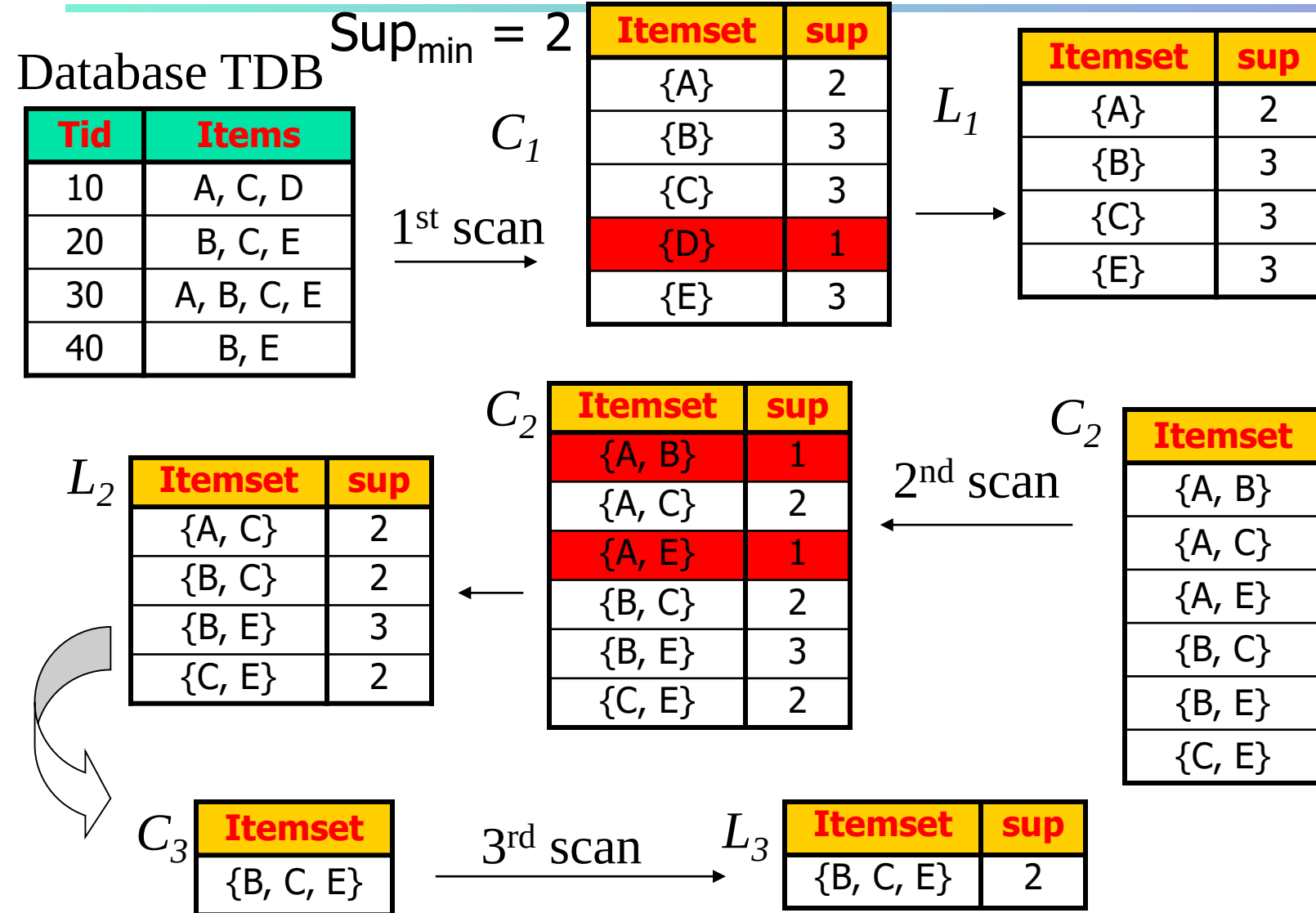
Step 1: Generate Frequent Itemsets

- **Set Minimum Support Threshold:** The user defines a minimum support threshold to identify which itemsets are considered frequent.
- **Generate Candidate Itemsets:** Start with the set of all individual items and count their occurrences in the dataset to form the first level of frequent itemset (L_1).
- **Iterative Process:** For each subsequent level k :
 - Generate candidate itemset C_k from the frequent itemsets L_{k-1} by joining them with themselves.
 - Count the support for each candidate itemset by scanning the database.
 - Retain only those candidates that meet the minimum support threshold to form the next level of frequent itemsets L_k .
- **Termination:** The process continues until no new frequent itemsets .

Step 2: Generate Association Rules

- Once the frequent itemsets are determined, the next step is to derive association rules from these itemsets. This involves:
- **Calculating Confidence:** For each frequent itemset, calculate the confidence for all possible rules that can be formed from that itemset.
- **Applying Confidence Threshold:** Retain only those rules that meet a predefined confidence threshold.
- **Evaluating with Lift:** Optionally, evaluate the rules using the lift metric to determine the strength of the associations.

The Apriori Algorithm—An Example



Generating the Association Rules

- $\{B,C,E\} \rightarrow 2$, Confidence = 75%

Rules	Support	Confidence	
$\{B,C\} \rightarrow E$	2	$2/2 = 1$	100%
$\{C,E\} \rightarrow B$	2	$2/2 = 1$	100%
$\{B,E\} \rightarrow C$	2	$2/3 = 0.66$	66%
$B \rightarrow \{C,E\}$	2	$2/3 = 0.66$	66%
$C \rightarrow \{B,E\}$	2	$2/3 = 0.66$	66%
$E \rightarrow \{B,C\}$	2	$2/3 = 0.66$	66%

Transactional Data for an *AllElectronics*
Branch

<i>TID</i>	<i>List of item_IDs</i>
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

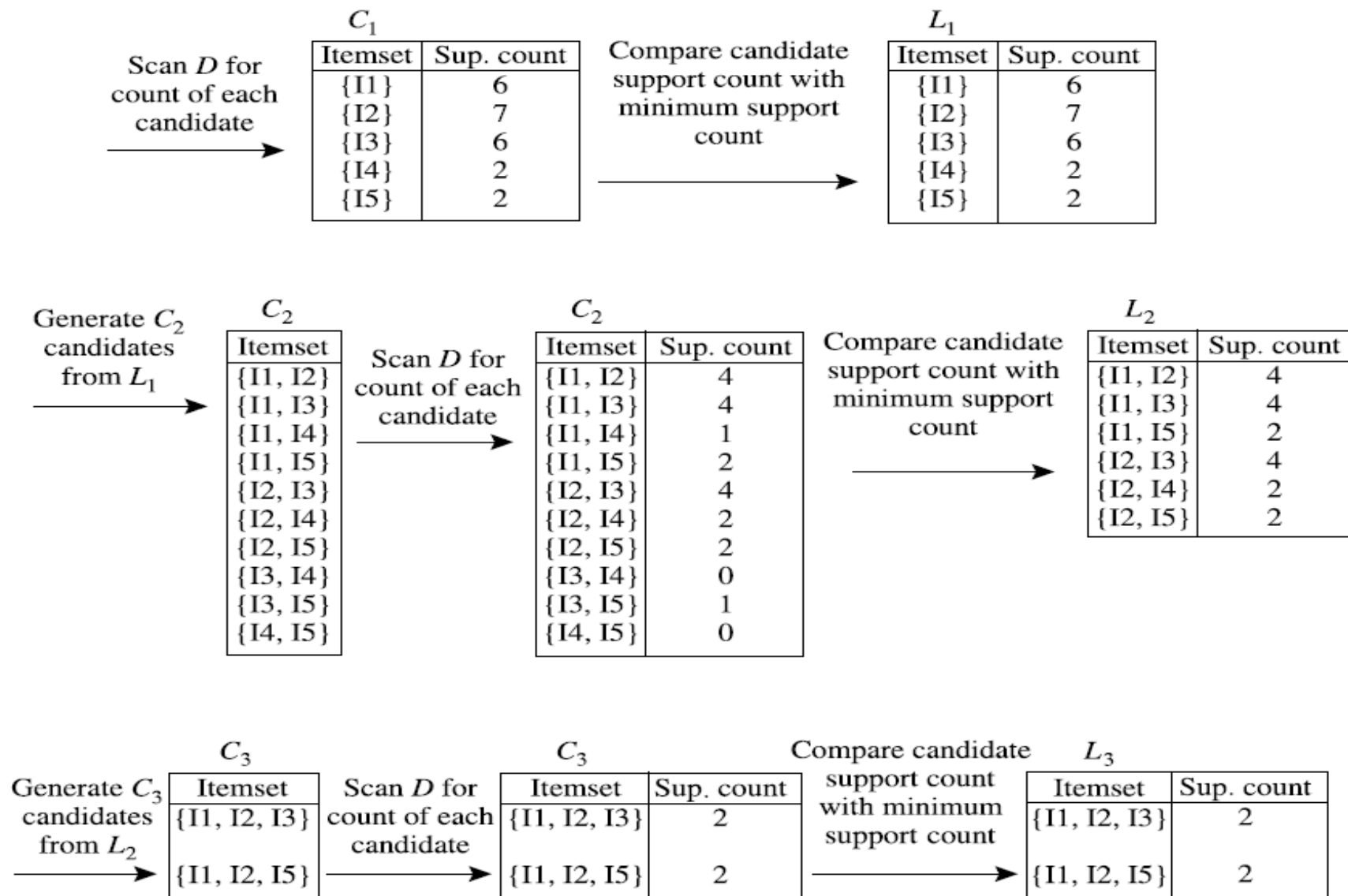


Figure 6.2 Generation of the candidate itemsets and frequent itemsets, where the minimum support count is 2.